



CS 540 Module Three Assignment Guidelines and Rubric

Overview

Understanding how to clean, prepare, and combine data is essential for working in data-driven environments. These skills are used in roles across industries, including public health, marketing, education, finance, and policy analysis. Professionals in these fields often face challenges with messy or incomplete data and must be able to structure and integrate information from multiple sources.

This assignment strengthens your ability to describe core data preprocessing techniques and explain how combining data sets can improve the quality and depth of analysis. These are foundational skills that support more advanced data modeling, visualization, and decision-making tasks later in your academic work and in professional settings. Accurate preparation and thoughtful data integration lead to stronger, more meaningful insights that inform real-world actions.

Directions

In this assignment, you will describe common data preprocessing techniques and explain how each contributes to effective data analysis. You will also analyze the importance of blending multiple data sources, including examples of how integrated data sets create more meaningful insights.

Specifically, you must address the following rubric criteria:

1. Describe multiple **data preprocessing techniques**.
2. Explain the **purpose and function** of each data preprocessing technique.
3. Explain the concept of **blending multiple data sets**.
4. Explain why **blending multiple data sets is important** in analysis and for making informed decisions.
5. Explain how you would apply data preprocessing and blending techniques in your own academic, professional, or personal **data work**.

What to Submit

Submit your assignment as a 2- to 3-page Microsoft Word document with double spacing, 12-point Times New Roman font, and one-inch margins.

AI Usage

If you use gen AI tools to support your work on this assignment, be sure to follow [these AI usage guidelines](#). You must acknowledge your use of these tools in your work. Guidelines on how to cite AI tools can be found in [this Shapiro Library guide](#).

Module Three Assignment Rubric

Criteria	Exceeds Expectations (100%)	Meets Expectations (90%)	Partially Meets Expectations (70%)	Does Not Meet Expectations (0%)	Value
Data Preprocessing Techniques	Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner	Describes multiple data preprocessing techniques	Shows progress toward meeting expectations, but with errors or omissions	Does not attempt criterion	19
Purpose and Function	Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner	Explains the purpose and function of each data preprocessing technique	Shows progress toward meeting expectations, but with errors or omissions	Does not attempt criterion	19
Blending Multiple Data Sets	Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner	Explains the concept of blending multiple data sets	Shows progress toward meeting expectations, but with errors or omissions	Does not attempt criterion	19
Importance of Blending Multiple Data Sets	Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner	Explains why blending multiple data sets is important in analysis and for making informed decisions	Shows progress toward meeting expectations, but with errors or omissions	Does not attempt criterion	19
Data Work	Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner	Explains how to apply data preprocessing and blending techniques in academic, professional, or personal data work	Shows progress toward meeting expectations, but with errors or omissions	Does not attempt criterion	19
Clear Communication	Exceeds expectations with an intentional use of language that promotes a thorough understanding	Consistently and effectively communicates in an organized way to a specific audience	Shows progress toward meeting expectations, but communication is inconsistent or ineffective in a way that negatively impacts understanding	Shows no evidence of consistent, effective, or organized communication	5
Total:					100%